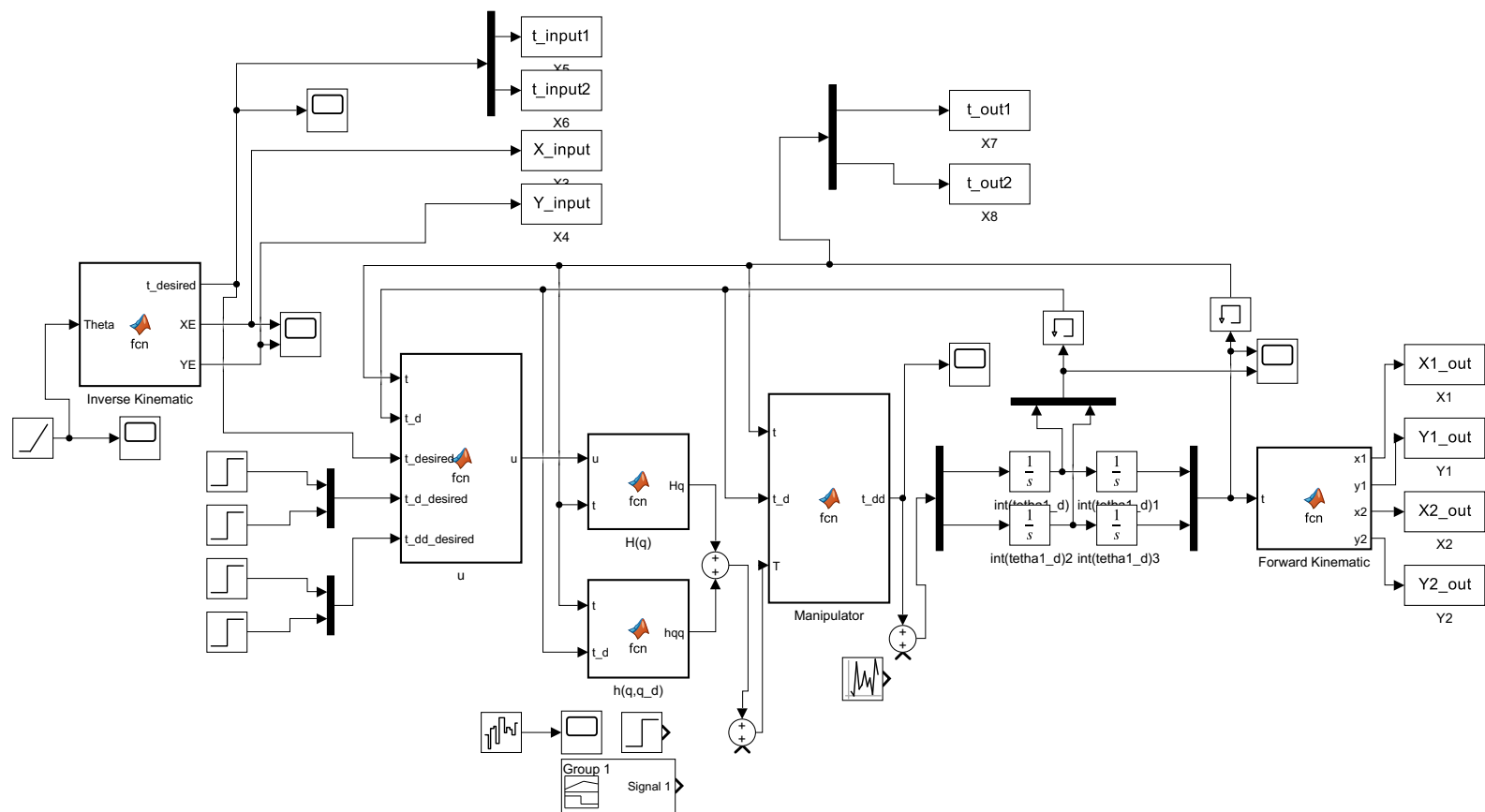




```
function [x1,y1,x2,y2] = fcn(t)
m1 = 1;
m2 = 1;
l1 = 1;
lc1 = 0.5;
l2 = 1;
lc2 = 0.5;
g = 9.81;
% Joint 2 location
x1 = l1*cos(t(1));
y1 = l1*sin(t(1));
% Joint 3 location
x2 = l1*cos(t(1))+l2*cos(t(1)+t(2));
y2 = l1*sin(t(1))+l2*sin(t(1)+t(2));
```

```
function Hq = fcn(u, t)
m1 = 1;
m2 = 1;
l1 = 1;
lc1 = 0.5;
l2 = 1;
lc2 = 0.5;
g = 9.81;

H11 = m1*lc1^2 + l1 + m2*(l1^2+lc2^2+2*l1*lc2*cos(t(2)))+l2;
H22 = m2*lc2^2 + l2;
H12 = m2*l1*lc2*cos(t(2)) + m2*lc2^2 + l2;
H21 = m2*l1*lc2*cos(t(2)) + m2*lc2^2 + l2;

H = [H11 H12; H21 H22];

Hq = H*u;
```

```

function [t_desired, XE, YE] = fcn(Theta)
l1 = 1;
l2 = 1;
% Trajectory Design (Circle)
XE = 0.8 + 0.6*cos(Theta);
YE = 0.8 + 0.6*sin(Theta);
if Theta >= 2*3.14-0.2 && Theta <= 2*3.14+0.2
    XE = 1.4;
    YE = 0.8;
end

if Theta > 2*3.14+0.2
    XE = 2;
    YE = 0;
end

%% Trajectory 2
% r = Theta/10;
% XE = 0.8+ r*cos(Theta);
% YE = 0.8+ r*sin(Theta);

%% Inverse Kinematic
t1_desired = 2*atan((2*YE*l1 + (- XE^4 - 2*XE^2*YE^2 + 2*XE^2*l1^2 + 2*XE^2*l2^2 - YE^4 + 2*YE^2*l1
t2_desired = 2*atan((2*YE*l1 - (- XE^4 - 2*XE^2*YE^2 + 2*XE^2*l1^2 + 2*XE^2*l2^2 - YE^4 + 2*YE^2*l1
% Output Theta
t_desired = [t1_desired; t2_desired-t1_desired];

```

```

function t_dd = fcn(t, t_d, T)
m1 = 1;
m2 = 1;
l1 = 1;
lc1 = 0.5;
l2 = 1;
lc2 = 0.5;
g = 9.81;
H11 = m1*lc1^2 + l1 + m2*(l1^2+lc2^2+2*l1*lc2*cos(t(2)))+l2;
H22 = m2*lc2^2 + l2;
H12 = m2*l1*lc2*cos(t(2)) + m2*lc2^2 + l2;
H21 = m2*l1*lc2*cos(t(2)) + m2*lc2^2 + l2;
H = [H11 H12; H21 H22];
h = m2*l1*lc2*sin(t(2));
g1 = m1*lc1*g*cos(t(1)) + m2*g*(lc2*cos(t(1)+t(2))+l1*cos(t(1)));
g2 = m2*lc2*g*cos(t(1)+t(2));
t_dd = inv(H)*(T - [-h*t_d(2) -h*t_d(1)-h*t_d(2); h*t_d(1) 0]*[t_d(1); t_d(2)]-[g1; g2]);

```

```
function hqq = fcn(t, t_d)

t1=t(1);
t2=t(2);
t_d1=t_d(1);
t_d2=t_d(2);

m1 = 1;
m2 = 1;
l1 = 1;
lc1 = 0.5;
l2 = 1;
lc2 = 0.5;
h = m2*l1*lc2*sin(t(2));
g = 9.81;

g1 = m1*lc1*g*cos(t1) + m2*g*(lc2*cos(t1+t2)+l1*cos(t1));
g2 = m2*lc2*g*cos(t1+t2);

hqq = [-h*t_d2 -h*t_d1-h*t_d2; h*t_d1 0]*[t_d1; t_d2]+[g1; g2];
```

```
function u = fcn(t, t_d, t_desired, t_d_desired, t_dd_desired)
```

```
KD = 20;
```

```
KP = 100;
```

```
tetha_tilted = t - t_desired;
```

```
tetha_d_tilted = t_d - t_d_desired;
```

```
u = t_dd_desired - (KD*(tetha_d_tilted)) - (KP*(tetha_tilted));
```